



OPEN Analysis and control of grid-interactive PV-fed BLDC water pumping system with optimized MPPT for DC-DC converter

J. Sevugan Rajesh^{1✉}, R. Karthikeyan² & R. Revathi³

In this study, a novel water pumping module fed by grid interactive Photo-Voltaic with a bidirectional Power Flow Control was proposed. In addition to improving the pumping system's reliability, a water pump is powered by a brushless DC motor drive. This method enables the pump to work at its maximum capacity for the entirety of that day, regardless of the weather. The entire system becomes more reliable as a result of the motor's increased use of photovoltaic (PV) generated power for pumping applications. Maximum Power Point Tracking (MPPT) controller incorporating Machine Learning algorithm drives bridgeless greater static gain DCDC converter to achieve higher power generation point and increment PV efficiency. The PV array's operation would be managed using the ML back propagation technology to capture the most electricity under any ecological circumstance. A BLDC motor is fed by a Voltage Source Inverter (VSI) that includes a DC bus controlled in both directions by a unit vector template (UVT) approach incorporated in a single-phase voltage source converter (VSC). Additionally, utilizing a PI controller to manage the DC capacitor voltage in the UVT controller at a particular level is not appropriate for the increased PQ capabilities. However, due to tuning problems with the current controller, this controller is unpopular. The aforementioned problems are resolved by employing a unique intelligent-based fuzzy logic controller that achieves good performance features. In this technique, the function of a PV array at its Maximum Power Point (MPP), as well as power quality enhancements and a decrease in Total Harmonic Distortion (THD) of the grid are accomplished. The proposed PI controller attains a significant voltage THD of 3.736. The PI controller, on the other hand, managed to achieve a load voltage THD of 2.629%. The ANFIS method, whose value is 1.739%, is discovered to have a lower THD than all remotes with improved features, it lessens abrupt swings while maintaining steady DC-link voltage.

Keywords Brushless DC motor, Voltage source inverter (VSI), Unit vector template (UVT), Maximum power point (MPP), Power quality, Fuzzy-logic controller (FLC), Total harmonic distortion (THD)

Instantaneous consumers are encouraged to use renewable energy as a result of the steadily rising carbon emissions and declining supply of fossil fuels¹. Solar PV generation is a promising replacement for traditional forms of generation for a variety of appliances^{2,3}. For home use, most contemporary industry, and agricultural use, groundwater pumping is a need. Regarding this, PV energy has been used as a supply to the pumping system in recent years⁴. This addresses the main issues of pumping water through a reliable source of energy and efficient resource usage. Conversely, DC motors have brushes and commentators that need frequent maintenance and are not suitable for pumping applications⁵. Due to the BLDC motor's, high power density, excellent efficiency, minimal maintenance, lengthy service life, little electromagnetic interference (EMI) concerns, and compact size, it has gained popularity during the past 10 years. Furthermore, to enhance the process and maintenance-free operation, it has been found that using this motor minimizes the PV panel's price and size^{6–12}. The solar panels can be connected to the DC motor-driven pumps through a boost converter to balance the impedance between the motor drive and the PV panel. Therefore, the above setup makes driving to the MPP simple. Inconsistency and nonlinear output characteristics in precise under shady conditions are the main flaws in solar-powered motor drives. Additionally, a lack of light (often at night) prevents the pumping unit from operating. For a

¹ECE Department, Dr. NGP Institute of Technology, Coimbatore, India. ²EEE Department, M. Kumarasamy College of Engineering, Karur, India. ³EEE Department, KPR Institute of Engineering and Technology, Coimbatore, India. ✉email: sevugan.rajesh123@gmail.com

PV-fed water pumping system to operate continuously and effectively, these restrictions must be shorted out¹³. The various kinds of literature^{14–18} make use of MPPT techniques to enhance the nonlinear SPV output characteristics, and an SPV-producing unit implemented for water pumping is connected to a utility grid to achieve consistent pumping operation to the fullest extent possible regardless of the weather¹⁹. PFC-based AC/DC-DC power converters are described in²⁰. Additionally, a power converter is necessary to convert the power into another readily available voltage level. Furthermore, as the current converters have high gain, switching losses, and a soft switching technique is needed^{21–25}.

A coupled inductor-based multi-input DC-DC converter is described in²⁶ its attain high gain with minimal losses. In PV-fed BLDC motor systems, choosing an appropriate DC-DC converter is therefore a crucial responsibility. To regulate the converter's output when MPPT systems are in use, a control strategy is also required²⁷. To achieve steady-state functioning in MPPT, a control strategy utilizing the PI controller is used. The discovery of optimization-based techniques is prompted by drawbacks like the peak overshoot problem. The nonlinearity and parametric fluctuation in load are increased by the traditional PI controller²⁸. Since they are straightforward to apply in real-time for constant environmental conditions and have a simple design, Incremental Conductance and conventional Perturbation and Observation (P&O)²⁹ are popular MPPT approaches. However, in rapidly changing climatic conditions, the P&O approach modifies the power output and, when many peaks occur, negatively impacts the system's stability. Particle Swarm optimization algorithm, firefly algorithm (FA), and Ant Colony Systems (ACS) are examples of nontraditional MPPT methodologies³⁰ and the capacity to handle numerous tips in the PV curve and accelerate convergence to the original global maximum position without power waste is a feature of various acknowledged and modern evolutionary algorithms (EA)³¹.

A Permanent Magnet Synchronous Motor is driven with a stand-alone PV system^{32–35}. A pumping system operated by a solar power-fed synchronous motor is also equipped with a two-stage energy conversion system³⁶. The PV is paired with a boost converter to increase output, which is optimized using the incremental conductance method. A PMSM-driven water pump with field-oriented control is also shown in³⁷. To obtain a superior MPPT operation, the controller will change the motor's reference speed. Due to its simple implementation and quick convergence, the ML back propagation technique is utilized in this paper. By locating and achieving the ideal Global Peak Point (GPP) under rapidly varying climatic conditions, the ML backpropagation algorithm successfully increases the system's efficiency. The solar PV fuelled pumping system that is connected to the grid is described in³⁸. An intelligent fuzzy-based high-gain DC-DC converter is described in³⁹. An effective hybrid grid-integrated solar system is generated in⁴⁰. Even though it is a grid-connected PV pumping system, it only receives power from and is controlled by the utility grid. The PV and grid-interactive system employing BLDC motor drive for pumping employs control of power flow in unidirectional⁴¹ in which at any time the necessary energy is obtained from the grid. When the pumping system is turned off, the aforementioned system is unable to utilize the energy from the PV. As a result, both the SPV setup and the pumping units are not being used to their full potential. All of these SPV-supported water pumping machines use the unidirectional flow of power, either drawing power from the grid or feeding it to the utility grid. Utilizing Pulse-Width Modulation (PWM) chopping techniques⁴² claims that a buck converter is employed to regulate the speed drive and reduce torque ripples. Switching losses in Voltage Source Inverters (VSIs)⁴³ will happen as a result of the high-frequency PWM signal; additionally⁴⁴ include a soft beginning buck-boost converter into BLDC water pumping. Ripples are thereby removed, and variable speed control is also accomplished⁴⁵.

To enhance the efficiency of the MPPT in scenarios where it works with the zeta converter, a few optimization approaches, like fuzzy and artificial bee colony algorithms, have recently been included⁴⁶. To monitor the Maximum Power Point (MPP) with fast convergence and achieve maximum power with minimal oscillations, an enhanced variable step size-radial Basis Function Network (RBFN)⁴⁷ in the NN method has been developed⁴⁸. There was a discussion of similar MPPT-based pumping devices in⁴⁹. A sensorless Sliding Mode Controller (SMC) is also constructed and used to implement a PV-supplied BLDC motor, as mentioned in⁵⁰.

The proposed approach employs a UVT-based modulation scheme to manage the enhancement of a bi-directional control, enabling the system to inject power from PV to the grid if pumping is in an off condition and from the single-phase grid to motor drives if the SPV power is insufficient (during the night) to employ the BLDC motor-pumping at its maximum capacity. Consumers can make money by selling electricity to the grid thanks to this enhanced capability. Due to their improved performance over PI controllers, the presented intelligent Fuzzy control systems are widely used in many applications^{51–55}. In this study, the effectiveness of a solar-PV and grid-integrated water pumping system is compared to that of a traditional PI and an intelligent fuzzy logic controller for PQ augmentation. This system uses the MATLAB/Simulink platform to operate at a PV array's MPP and improve power quality in other ways, such as PFC and a reduction in the grid THD. The grid power quality requirements for power transmission are maintained as per the IEEE-519 standard.

A bidirectional power flow controller refers to a system capable of managing the flow of electrical energy in both directions between the grid and the photovoltaic (PV) system. This allows the water pumping system to utilize energy from the grid when solar power is insufficient (for example, during nighttime or cloudy conditions) and enables surplus energy generated by the PV system to be fed back into the grid when the pumping system is not in operation. This approach enhances the overall efficiency and reliability of the water pumping system by ensuring that it can operate continuously regardless of environmental conditions.

The primary objective of our research is to develop an efficient and reliable water pumping system that maximizes energy utilization from solar PV sources while maintaining power quality. The main contributions of this work are.

- a. Developing a UVT-based modulation scheme for a bi-directional control between grid and solar PV system.
- b. Developing an optimized MPPT converter that is suitable for integrating solar PV with a grid with the support of intelligent fuzzy systems.

Proposed water pumping system

Figure 1 shows a construction of the recommended system of water pumping which is powered by a BLDC motor. A step-up converter, VSI, and a PV together feed a BLDC motor-pumping system. The step-up converter uses the incremental conductance algorithm to execute MPPT on the PV array, while the VSI manages the BLDC motor electronic commutation⁵⁶. To perform an electrical commutation, an embedded encoder produces three Hall-Effect signals. A single-phase utility grid supports the VSI DC bus. Through a DC bus capacitor, a Voltage Source Converter (VSC)⁵⁷ makes it possible to transfer power in both directions. PV supplies grid only if a water pump is not necessarily else, power can flow between the VSC and grid with the help of an interface inductor, which is also used to minimize the harmonic current entering the supply. To reduce the harmonics on supply voltage, an RC ripple filter is offered.

Water-pumping system using UVT-ANFIS controller fed BLDC motor

The phase current sensors that were introduced previously are removed by the planned BLDC motor drive. Irrespective of weather conditions, the BLDC-incorporated water pump is required to run at rated speed. Subsequent control of DC bus voltage and working speed, enables a bi-directional PFC making it possible to convey the power needed to pump the water at maximum capacity. The integration of a Brushless DC (BLDC) motor significantly improves the efficiency of the water pumping process in several ways like Improved Power Density, Precise Speed Control, Lower Electromagnetic Interference, and Reduced Maintenance. Overall, the integration of a BLDC motor leads to a more efficient, reliable, and adaptable water pumping system, making it a superior choice for renewable energy applications. Figure 2 shows the Simulink model of the proposed system.

If there is no grid, the BLDC motor's rated DC voltage is not maintained under adverse weather circumstances, and the changeable DC bus voltage controls the speed. To fully utilize the available resources, a grid-interactive PV generation must be able to produce a dependable water pumping system. As shown in Fig. 3, a UVT generation-based bi-directional power regulation is employed to allow power to flow in either direction. Given that it doesn't require a complicated mathematical model or procedure, this is the simplest technique and one of the easiest to use. A Phase Locked Loop (PLL) is utilized to synchronize the grid voltage and current. On the contrary, by controlling the DC bus voltage (V_{dc}), a magnitude of the essential supply current, I_{sp} , is obtained. The voltage is regulated via a proportional-integral (PI) controller. To reduce the ripple contents, V_{dc} is detected and processed through a first-order low-pass filter. Next, the filtered V_{dc} is contrasted with a predetermined value, V_{dc}^* . I_{sp} and $\sin$ are multiplied to obtain an essential element of supply current, i_s^* . To produce the gating pulses for VSC, a current controller compares the sensed supply current, i_s , with i_s^* , and processes any errors. The voltage regulator produces a positive I_{sp} when a utility power draw is necessary. As a result, the grid is used to supply current that is in phase. Similar to this, an out-of-phase current results from a negative I_{sp} generated when a grid is fed by a PV. So, the direction of power flow can be adjusted to meet requirements by flipping the direction of the current. The DC bus voltage can't be controlled if there's no grid available. Although susceptible to weather conditions, the PV array may still power the water pump in independent mode. The system effectively

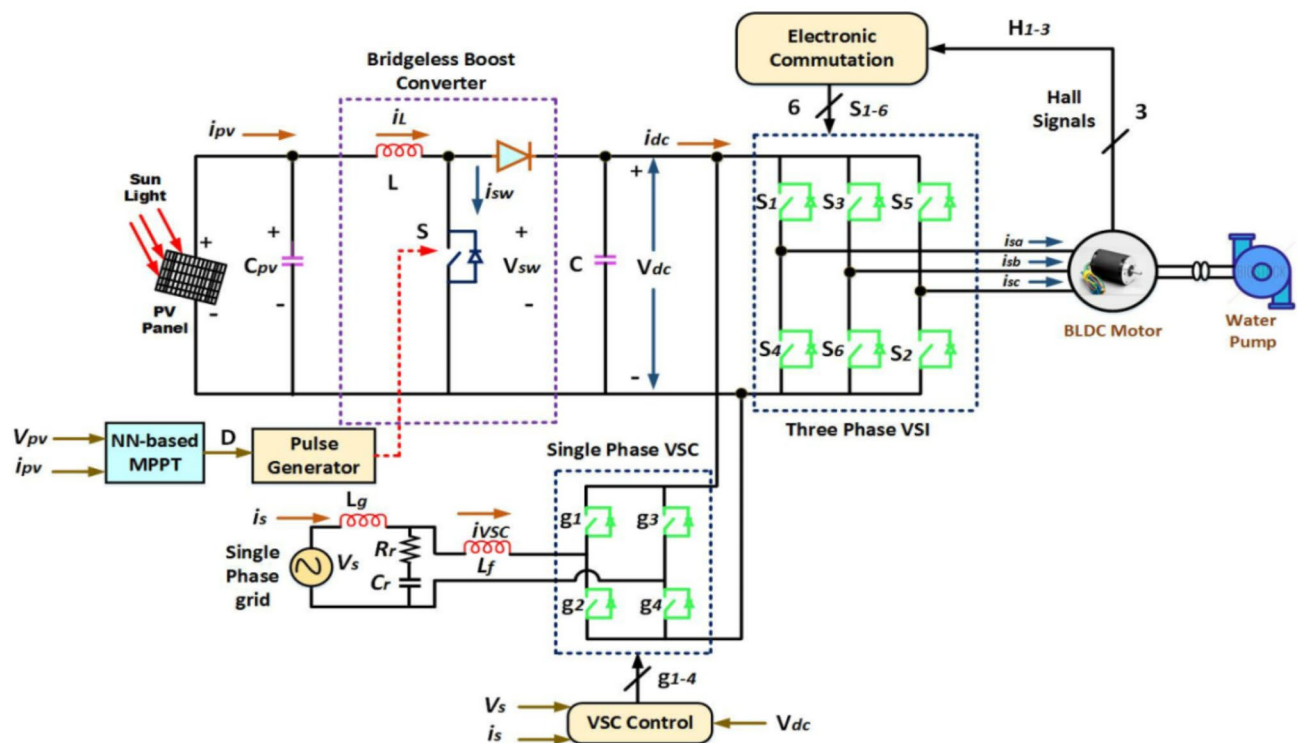


Fig. 1. Grid and PV-based water pumping system powered by a BLDC motor.

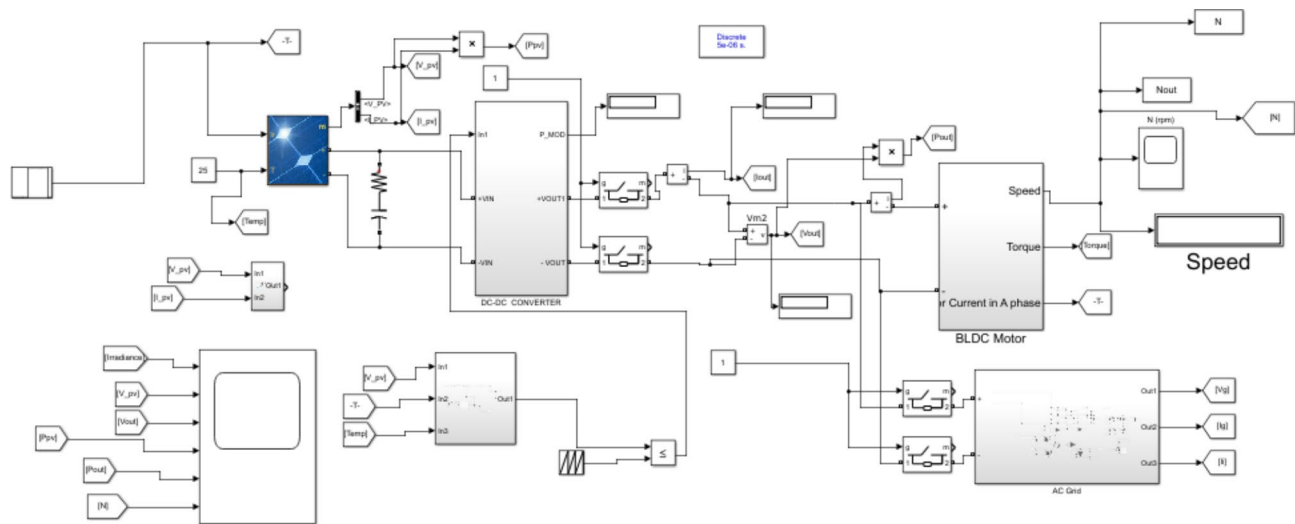


Fig. 2. Simulink model of the Grid and PV-based water pumping system powered by a BLDC motor.

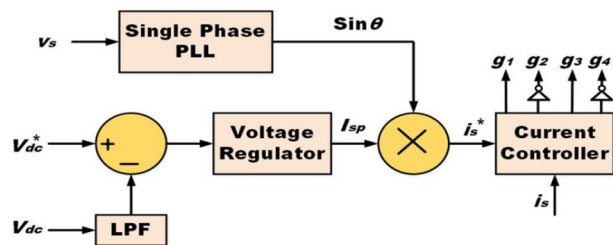


Fig. 3. Control of the VSC's bidirectional power flow based on UVT.

Converter	Category			Control	Voltage & power	Capacitors + Inductors	Semiconductor
	BBC	SBBC	BLBC				
Conventional	✓	✓	✓	Linear & Non-linear	Low Voltage & power	BBC-2 SBBC-2 BLBC-2	BBC-6 SBBC-4 BLBC-4
Interleaved	✓		✓	Linear & Non-Linear	Low voltage & power	BBC-3 BLBC-5	BBC-8 BLBC-6
Bidirectional	✓			Non-linear	Low Voltage & power	BBC-6	BBC-3
Soft switching		✓		Linear & Non-linear	Low voltage	SBBC-5	SBBC-6
Clamped diodes and switches		✓		Non-linear	Low voltage & power	SBBC-3	SBBC-8
Conventional GaN			✓	Non-linear	Low Voltage & power	BLBC-2	BLBC-4
Pseudo totem-pole			✓	Non-linear	Not recommended	BLBC-3	BLBC-6

Table 1. Comparison among BBC, SBBC, and BLBC topologies categorized.

manages power flow during varying weather conditions using Bidirectional Power Flow Control, Maximum Power Point Tracking, Intelligent Fuzzy Logic Controller, Energy Storage Utilization, and Adaptive Operation.

Comparative analysis of bridge and bridgeless boost converter

Due to their general simplicity, Bridge Boost Converter (BBC) topologies have been utilized recently for powering DC loads. Achieving high PF and low THD levels is possible using BBC. The following comparative drawbacks are present in this architecture when employed in medium and high-power applications. (1) Conduction losses at the Bridge Boost Converter are increased by the application of at least three switches along the current's route from supply to load, (2) To reduce the THD below the IEC 1000-3-2 standard, BBC needs a switching frequency larger than 30 kHz.; (3) Design concerns are needed because the inductor is on the DC side and must prevent core saturation. Only applications up to 1 kW are advised for use with BBC topology in its standard version^{57–59} In a BBC rectifier, the power used by the line bridge rectifier can account for 30 to 60% of total loss across a

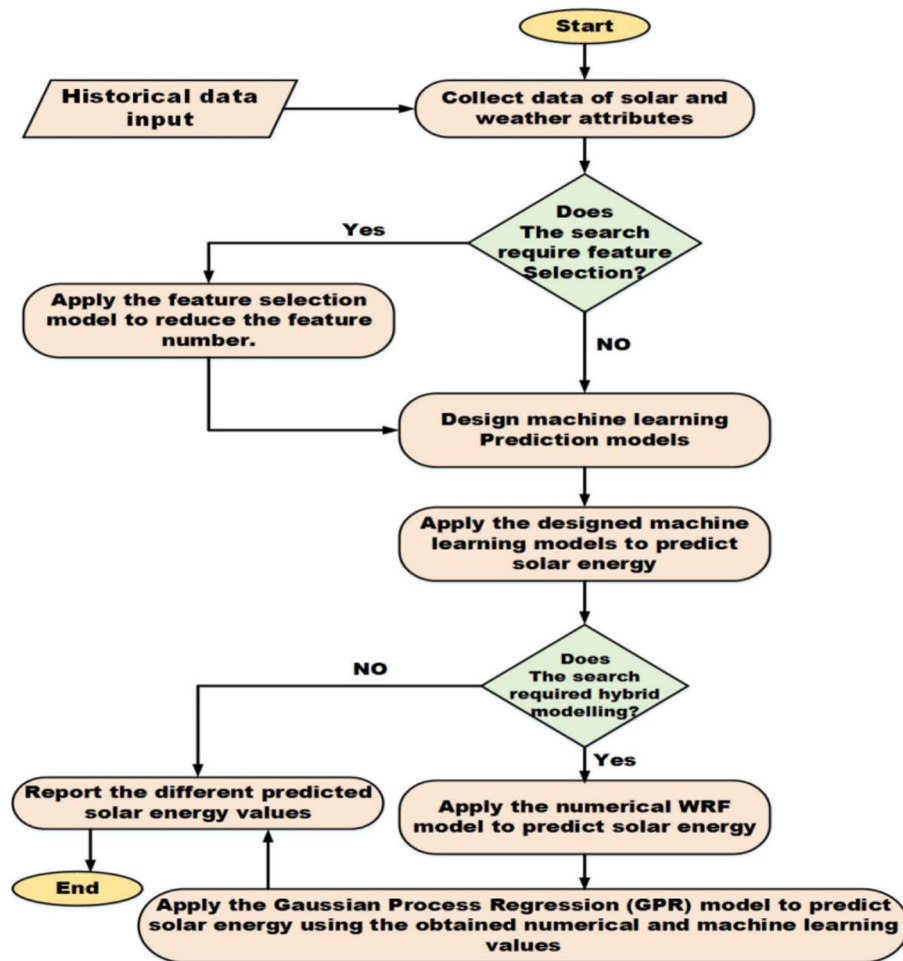


Fig. 4. The proposed solar prediction models' flowchart.

wide range³⁰. Reduced conduction losses are possible with Semi-Bridgeless Boost Converter (SBBC) topologies because, in comparison to BBC topologies, SBBC topologies have fewer semi-conductive switches along the source to load the current path. Furthermore, the input current of this converter has high common mode noise, necessitating additional elements and making the circuit more complex. The floating output ground, which pulses in time with the switching frequency, is what causes the common mode problem. Additionally, this design calls for an additional inductor, which raises the power converter's mass, complexity, and expense. By including diodes, capacitors, and inductors in this topology, as well as placing symmetrical switches for the phase and neutral lines in the right places, EMI issues at the common coupling point can be minimized⁶⁰.

Reverse-recovery currents from diodes are lessened by the asymmetrical placement of Bridgeless Boost converter (BLBC) components in the converter branches. Every half-line cycle, diodes clamp the output voltage to the input, it is free of the common mode interference issue. Furthermore, this architecture may be capable of bidirectional power conversion. However, because of the inherent asymmetry, every branch drive must be isolated, which adds to the controller's complexities^{61–65}. A comparative analysis of bridge and bridgeless boost converters is illustrated in Table 1.

Solar radiation prediction method

The primary component of the efficient design and operation of solar power systems is data on solar radiation⁶⁶. The stability of the electricity provided by solar power stations must be guaranteed. Hence, from an operations perspective, it is crucial to anticipate the amount of solar radiation at a certain place. However, achieving such a goal has some real-world challenges. The PV is specifically constrained by inaccurate predictions of solar radiation levels compared to various possibilities.

Numerous prediction models, such as artificial intelligence models and numerical weather prediction (NWP), have been proposed in the literature to address this issue. e.g.^{67–69}. , . To be sure, there are a lot of variables involved in the forecast process, such as topographic and weather variables, which have a big impact on the underlying prediction models. The model for solar forecast according to the flow of the proposed is presented in Fig. 4. To increase predictor performance and lower real-time prediction systems computing costs. Along with NWP models may supply predictions of solar radiation several days in advance⁷⁰. The operation of solar plants can benefit from the optimization of such information. A mesoscale model is an NWP model that scales

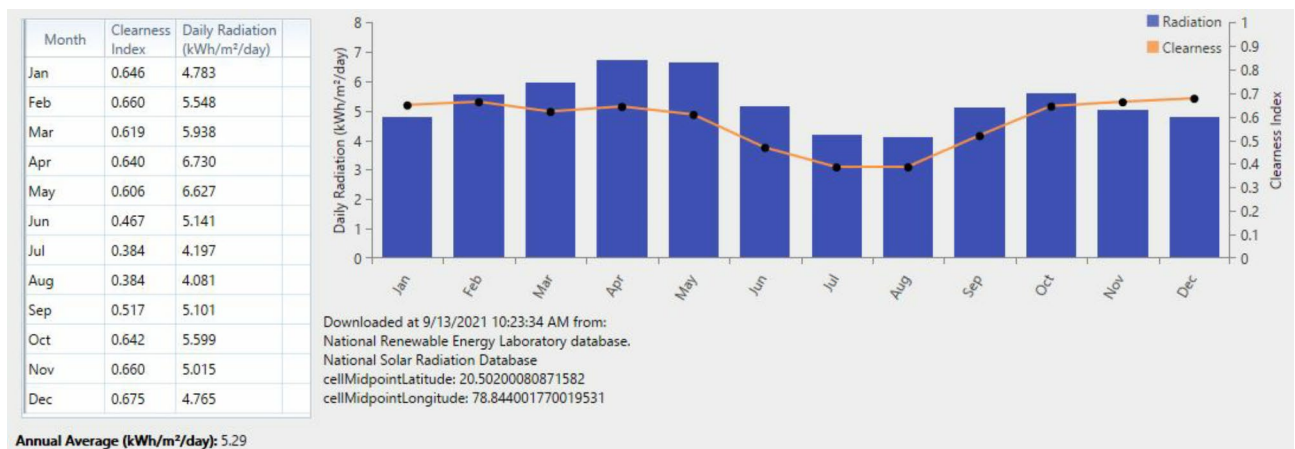


Fig. 5. Predicting monthly solar radiation at Vellore.

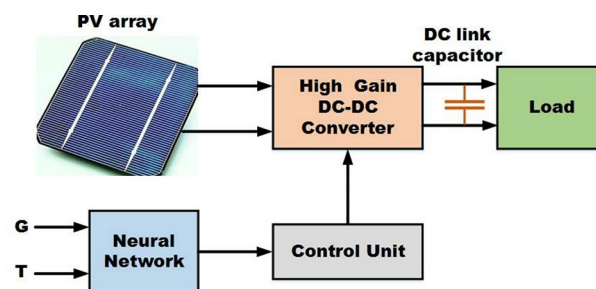


Fig. 6. PV and its control modules are included in the high-gain DC-DC converter.

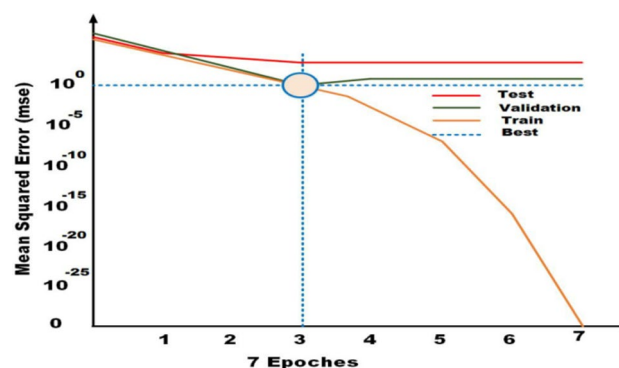


Fig. 7. Performance of neural network.

down reanalysis data. Mesoscale models contain more information than global-scale models since they operate across a smaller geographic area. As a result, these models can produce solar irradiance estimates with great temporal spatial resolution across a large area using higher amounts of computational power. The prediction of solar energy requires a lot of data, which calls for a lot of measurement tools and equipment. Additionally, it is frequently computationally expensive to calculate the weather data required for the projection procedure⁷¹. The reduction of data reading and computation processes necessary for energy forecast, as well as the cost of this process, are therefore two of the research's primary reasons. This contributes to the broadening and expansion of prediction activities, even beyond the purview of conventional measuring stations^{72–75}. Utilizing the potential of intelligent and hybrid systems to forecast short-term solar energy levels is another important driving force for this work. As a result, for investigation, the feature extraction method offered by a modified Tabu Search Attribute Reduction (TSAR)⁷⁶ is utilized along with the level of solar radiation prediction employing different methods of prediction. Figure 5 shows the performance of the predicting solar radiation for the given monthly. Figure 6 shows the PV and its control modules are included in the high-gain DC-DC converter.

ML-based MPPT control scheme

In this research work, Artificial Neural Networks (ANN) and Particle Swarm Optimization (PSO) are used as ML-based MPPT controllers. Here ANN are mathematical models that replicate the neural networks seen in the human brain and are used to tackle challenging contemporary issues. An ANN is made up of many layers of tiny processing units known as neurons. The neuron performs information gathering and output production, respectively. The application of ANN comprises a discussion of the theory, learning principles, and implementations of every major machine learning algorithm, as well as definitions and computation methods. The ANN model defines the terms activation function, neurons, inputs, and outputs. To adjust weights as parameters, change, the ANN then aids in deciding what kind of learning should be employed ANN training, and quality prediction. This paper implies ANN to achieve the MPP. The two variables used as inputs are irradiance (G) and temperature (T), while the output is the MPP voltage. The neuron weights in different layers are as a consequence calculated. PV model programming is used to collect data in MATLAB. Different methods can be implemented to train ANN, here, the backpropagation approach controls the PV operation to collect the most electricity possible. Levenberg-Marquardt is the algorithm in use.

Ten of the 16 data sets are utilized for training, four for validating, and two for testing. The goal of the Back Propagation (BP) algorithm training procedure is to minimize the mean square error between the feed-forward net's actual and expected outputs. The BP algorithm's error function is as follows:

$$Error = V_{REF} - V_{OUT} \quad (1)$$

A high gain boost converter and a control unit powered by a neural network make up Fig. 7. To operate a higher gain power converter with a predetermined V_{mpp} and I_{mpp} duty cycle at any given moment, use the following expression:

$$D = 1 - \sqrt{\frac{V_{mpp} * I_{out}}{I_{mpp} * V_{out}}} \quad (2)$$

The performance of the neural network is illustrated in Fig. 7.

Bi-directional PFC

A grid and PV generation enables the creation of a dependable water pumping configuration and the maximum utilization of the available resources. UVT-based bidirectional PFC⁷⁷ is used, as illustrated in Fig. 3, to permit the flow of power in either direction. Given that it doesn't require a complicated mathematical model or procedure, this is the simplest technique and one of the easiest to use. Using a single-phase PLL, the grid voltage and current are synchronized at fundamental frequency. It developed a sinusoidal supply voltage unit vector, $\sin \theta$. On the other hand, controlling the DC bus voltage, (v_{dc}), allows for the extraction of the amplitude of the supply current's foundational element, I_{sp} . The voltage is regulated via a PI controller. To eliminate the contents of the ripple, v_{dc} is measured and routed through a first-order low-pass filter. After that, the filtered v_{dc} is contrasted with V_{dc}^* , a predetermined value. I_{sp} and $\sin \theta$ are multiplied to obtain an essential element of supply current, i_s^* . To produce the gating pulses for VSC, a current controller compares the measured supply current, i_s , with i_s^* , and processes any errors. The voltage regulator produces a positive I_{sp} when it is necessary to draw electricity from a utility. As a result, the grid is used to draw an in-phase supply current. Similarly, when a PV is utilized to supply the grid, a negative I_{sp} is created, which causes the source current to be out of phase. Therefore, the power flow direction may be changed to suit the situation by flipping the direction of the current. An enhanced power quality at the grid is also secured by the implemented control technique. If the grid is absent, it is impossible to manage the DC bus voltage. Despite being climate-sensitive, the PV can still power the water pump standalone⁷⁸.

Proposed bidirectional PFC analysis

Figure 2, the currently controlled VSC is utilized to manage the V_{dc} and synchronize the grid by blocking the current control loop and the voltage control loop. Figure 8 emphasizes the block diagram of the proposed PFC technique. Implementing voltage and current compensators (G_{V_PID} and G_{I_PID}) that are stable and dynamic is the major goal.

The input-current transfer function's control is secure as⁷⁹

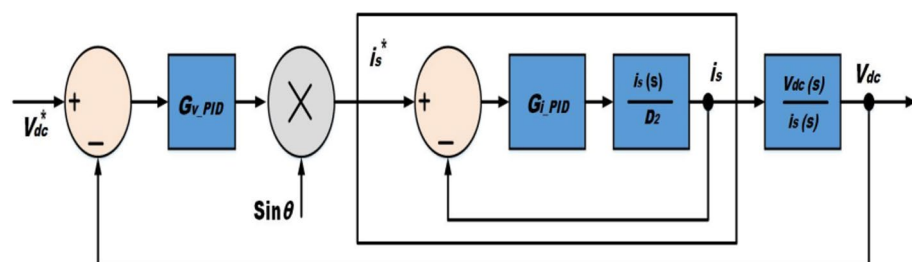


Fig. 8. Block diagram of proposed power flow control technique.

$$\frac{i_s(s)}{D_2} = \frac{V_{dc}Z_r}{Z_gZ_r + Z_f(Z_g + Z_r)} \quad (3)$$

where

$$Z_f = r_f + L_f s \quad (4)$$

$$Z_g = r_g + L_g s \quad (5)$$

$$Z_r = r_r + 1/C_r s \quad (6)$$

The r_f and r_g are L_f and L_g the internal resistances, accordingly.

The PID (Proportional-Integral-Derivative) controller, which is widely utilized for control loop compensation, is characterized as,

$$G_{i_PID} = K_{ip} + \frac{K_{ii}}{s} + K_{id}s \quad (7)$$

Therefore, the current loop gain can be calculated as,

$$G_{i_PID} \cdot \frac{i_s(s)}{D_2} = \left(K_{ip} + \frac{K_{ii}}{s} + K_{id}s \right) \cdot \frac{i_s(s)}{D_2} \quad (8)$$

The energy balance equation, which yields the following result, is used to determine the open loop gain.

$$E_{dc} = E_g + \frac{1}{2} C v_{dc}^2 \quad (9)$$

where E_{dc} - the input energy to the DC-link, E_g - the energy expressed by the grid and $(1/2) C v_{dc}^2$ - the power a DC link capacitor stores.

Under the assumption that the converter is lossless, the average power balance equation obtained by differentiating (9) by time can be used to determine the relationship between variations in the V_{dc} and the fundamental grid current.

$$P_{dc} = P_g + \frac{d \left(\frac{1}{2} C v_{dc}^2 \right)}{dt} \quad (10)$$

When examining the link between specific variables systems, other factors that have the least influence and contribution to the variables under research might be largely excluded to provide a more straightforward sensitivity analysis. As a result, P_{dc} is disregarded when analyzing grid current magnitude changes that affect the average V_{dc} ⁸⁰. If PV power is assumed to be zero, then $P_{dc}=0$, the only factor affecting the DC link capacitor.

$$-P_g = \frac{d \left(\frac{1}{2} C v_{dc}^2 \right)}{dt} \quad (11)$$

$$-\frac{v_s^\wedge i_s^\wedge}{2} = \frac{d \left(\frac{1}{2} C v_{dc}^2 \right)}{dt} \quad (12)$$

Applying slight deviations from the operational point causes:

$$\frac{d \left[\frac{1}{2} C (v_{dc} + \Delta v_{dc})^2 \right]}{dt} = -\frac{v_s^\wedge (i_s^\wedge + \Delta i_s^\wedge)}{2} \quad (13)$$

The result of disregarding steady-state values and the square of smaller perturbations is,

$$\frac{d \left[\frac{1}{2} C \times 2 v_{dc} \Delta v_{dc} \right]}{dt} = -\frac{(v_s^\wedge + \Delta i_s^\wedge)}{2} \quad (14)$$

The necessary outer voltage control loop transfer function is produced by rearranging the equations in (14) as,

$$C v_{dc} S v_{dc}(s) = -\frac{v_s^\wedge i_s^\wedge(s)}{2} \quad (15)$$

$$\frac{v_s(s)}{i_s(s)} = -\frac{v_s^\wedge}{2s C v_{dc}} \quad (16)$$

According to the description of the PID controller utilized for voltage control loop correction,

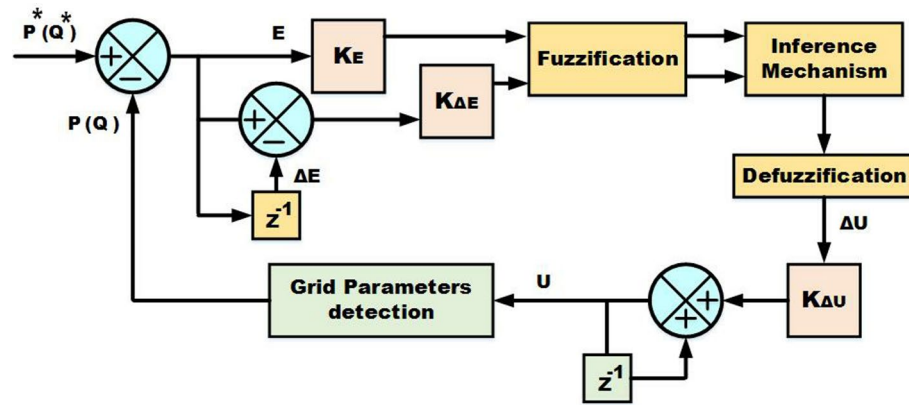


Fig. 9. The fuzzy controller diagram.

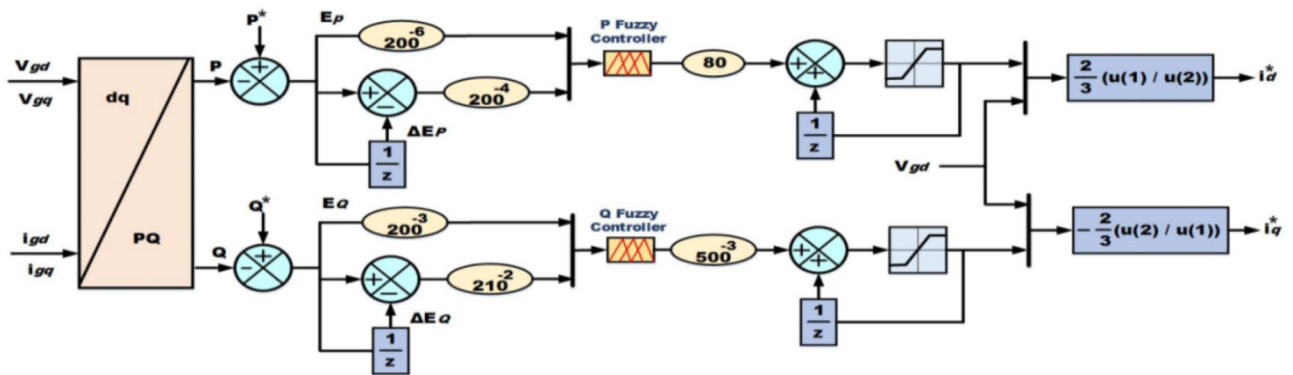


Fig. 10. PQ fuzzy logic control diagram.

$$G_{v_PID} = K_{vp} + \frac{K_{vi}}{2} + K_{vd}S \quad (17)$$

As a result, the voltage loop gain is calculated as,

$$G_{v_PID} \cdot \frac{v_{dc}(S)}{i_s(s)} = - \left(K_{vp} + \frac{K_{vi}}{s} + K_{vd}S \right) \times \frac{100.31}{S} \quad (18)$$

Model of PQ adaptive neuro-fuzzy inference system

In contrast to other common control methods, the adaptive neuro-fuzzy inference system (ANFIS) does not necessitate a prior understanding of the system's statistical model. However, an intelligence control technique requires a deposit of guidelines that are mostly dependent on the system⁸¹. The logic fuzzy controller (FLC) description demonstrates that the error and change of error of active and reactive power are important variables for control. The PV array supplies the active power reference P^* , while the reactive power reference Q^* is set to zero. The two characteristic parameters designated as E and ΔE may be utilized as the FLC inputs. The fundamental method of FLC is an⁸². In Fig. 9, the FLC diagram is illustrated and PQ FLC diagram is illustrated in Fig. 10. It is made up of 4 blocks that are described as follows:

- The error E , its change in error ΔE , and the fluctuation of the changeable control are connected to the normalization factors.
- The error's block of fuzzing and variations on it.
- The FLC rules.
- The FLC variation is converted into a numerical value using the defuzzification block^{83–85}. PQ fuzzy rules are mentioned in Table 2.

The proposed control strategy for $E(k)$, $\Delta E(k)$, and the rule viewer utilizing a membership FLC is shown in Fig. 11.

Variation of error (ΔE)								
		NB	NM	NS	ZE	PS	PM	PB
Error (E)	NB	NB	NB	NB	NM	NB	NS	ZE
	NM	NB	NM	NM	NM	NS	ZE	PS
	NS	NB	NM	NS	NS	ZE	PS	PM
	ZE	NB	NM	NS	ZE	PS	PM	PB
	PS	NM	NS	ZE	PS	PS	PM	PB
	PM	NS	ZE	PS	PM	PM	PM	PB
	PB	ZE	PS	PS	PM	PB	PB	PB

Table 2. PQ fuzzy rules.

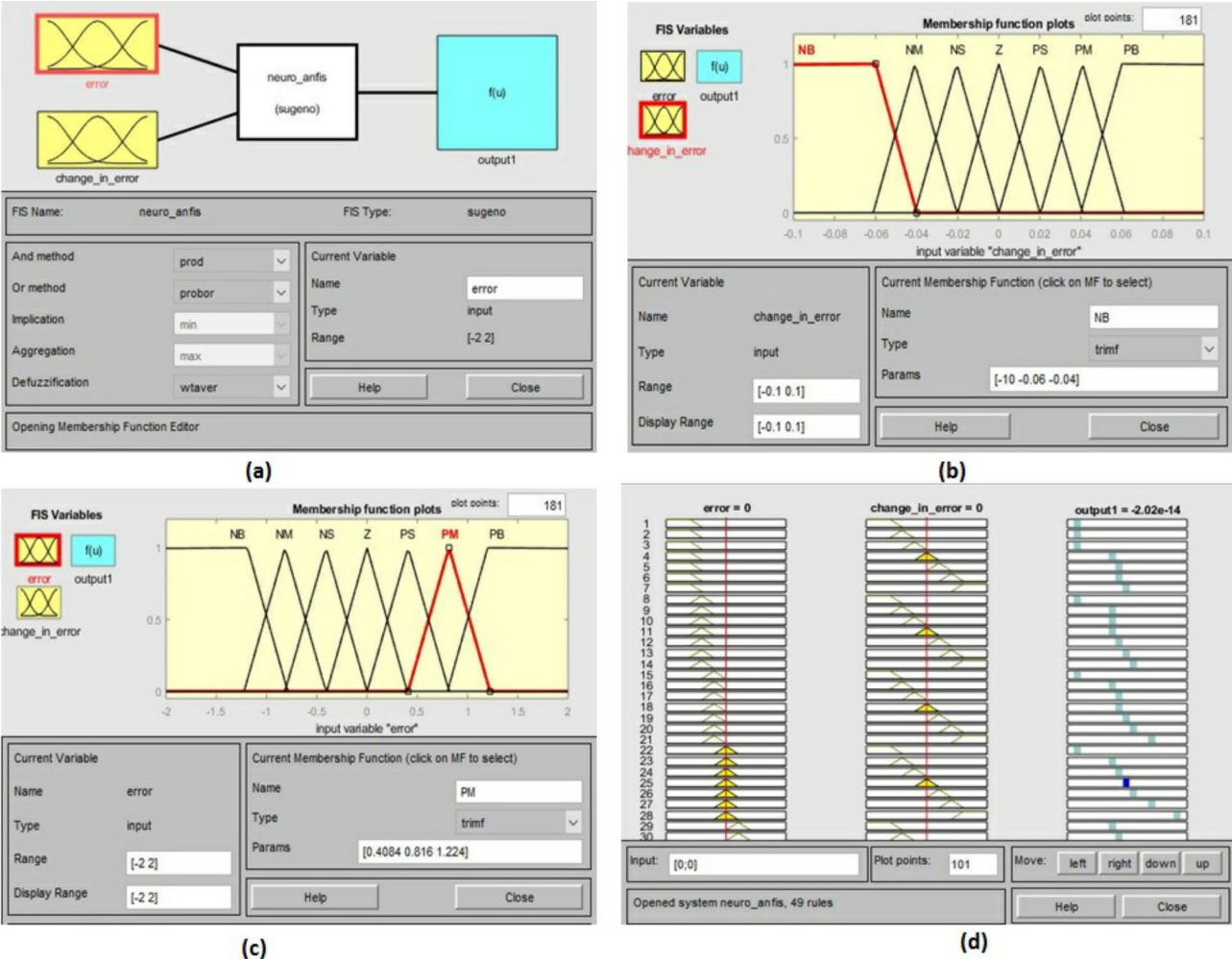


Fig. 11. (a) Design of fuzzy (b) Input variable DE (k) Membership functions (c) Input variable E (k) Membership functions (d) Rule viewer.

Design consideration Design of PV array

A 1.3 kW-BLDC motor pump is intended to be driven by a 1.5 kWp-PV array. It accounts for the power losses brought on by the motor pump and the converters. At the typical conditions (1000 W/m², 25 °C, AM 1.5), the parameters are estimated. To create a PV array with the requisite capacity, a PV module-BMU/214 with an MPP voltage and current of 28.5 V and 7.5 A⁸⁶ is used. First, the VSI DC bus voltage is taken into consideration while choosing the PV voltage at MPP. The following values are calculated following the chosen value of $V_{mpp} = V_{pv} = 200$ V as the current at MPP,

$$I_{mpp} = i_{pv} = \frac{P_{pv}}{V_{pv}} \quad (19)$$

where $P_{mpp} = P_{pv} = 1500 \text{ W}$ is the PV power at MPP.

Series and Parallel modules are as,

$$N_s = \frac{V_{mpp}}{V_m} \quad (20)$$

$$N_p = \frac{I_{mpp}}{I_m} \quad (21)$$

where V_m and I_m are voltage and current at the MPP of a module.

Modules are connected in a series to provide the necessary size PV array, as shown in (20) and (21).

Design of bridgeless boost converter

The input inductor L , is required for the design of a boost converter. It has been chosen so that, irrespective of the weather, the converter will run in CCM. Estimates for the duty ratio, D_1 , are,

$$D_1 = \frac{V_{dc} - V_{pv}}{V_{dc}} \quad (22)$$

where V_{dc} is the VSI- DC bus voltage.

The inductor, L is calculated as,

$$L = \frac{D_1 V_{pv}}{f_{sw} \Delta I_L} \quad (23)$$

where f_{sw} is the switching frequency; ΔI_L is a ripple in the current through the inductor.

Three-phase VSI design

The BLDC motor is supplied with three phases by a three-phase VSI. Voltage, current, and VA ratings are estimated during the design phase. The needed voltage value of an IGBT is determined using the V_{dc} of 270 V as the starting point.

$$V_{VSI} = V_{dc} \times 1.4 \quad (24)$$

To account for the switching transients, a voltage safety factor of 1.4 is chosen. Similar to this, the current rating of switches' is determined as follows:

$$I_{VSI} = I_{dc} \times 1.3 \quad (25)$$

where 1.3 – a factor of the current safety factor.

Finally, it is predicted that the needed VSI VA rating will be,

$$VA_{VSI} = V_{VSI} \times I_{VSI} \quad (26)$$

Single phase VSC design

The direction of a bidirectional power flow is managed by a single-phase VSC. The V_{dc} determines the switching devices' blocking voltage in a single-phase VSC. The switches must block the 270 V V_{dc} because that is what it is. A safety factor of 1.4 is used to account for voltage transients brought on by high switching frequency. Consequently, the IGBT devices' anticipated rating of voltage is as follows:

$$V_{VSC} = V_{dc} \times 1.4 \quad (27)$$

The maximum current that can be fed into or extracted from the grid by the VSC. The said current is calculated to be,

$$I_{s, max} = \sqrt{2} \frac{P_{mpp}}{V_s} \quad (28)$$

where $V_s = 180 \text{ V}$ - rms voltage value of the grid.

Consequently, IGBT devices can handle a maximum current of 11.78 A. The current rating is assessed to be, using a safety factor of 1.3, as follows:

$$I_{VSC} = V_{s, max} \times 1.3 \quad (29)$$

Therefore, it is predicted that VSC needs a VA rating of

$$VA_{VSC} = V_{VSC} \times I_{VSC} \quad (30)$$

Design of common DC link capacitor

The step-up converter's three-phase VSI and single-phase VSC share a capacitor. The single-phase grid voltage's second harmonic, which can be seen on the single-phase VSC's DC bus, is the harmonic that the device is tuned for. Therefore, it is estimated that the capacitor is,

$$C = \frac{I_{dc}}{2\omega_L \Delta V_{dc}} \quad (31)$$

where I_{dc} - regular current going via the DC bus, ω_L - frequency of the line in rad/s, and ΔV_{dc} - a ripple in the V_{dc} .

Interfacing inductor design

The allowed current ripple, ΔI_{VSC} , affects the choice of an interface inductor, L_f . According to estimates,

$$L_f = \frac{mV_{dc}}{4a f_{SW} \Delta I_{VSC}} \quad (32)$$

where modulation index, $m=1$, $\Delta I_{VSC} = 20\%$ of I_{VSC} , switching frequency, $f_{SW} = 10$ kHz, current ripple, overloading factor, $a = 1.2$.

R-C ripple filter design

The harmonic switches produced by the VSC are reduced by utilizing a first-order high-pass filter. On the utility grid side, a simple R-C filter is connected. This ripple filter's construction delivers very low impedance to the switching frequency elements and very high impedance to the fundamental frequency. For this requirement to be satisfied, $R_f C_r < T_{sw}$; where R_f - the ripple filter resistance, C_r - ripple filter capacitance, and T_{sw} - switching time.

C_r is estimated as,

$$C_f = \frac{T_{sw}}{4R_f} \quad (33)$$

To achieve this, a series configuration of 5 Ω , resistance, and 5 F capacitance is chosen as the RC ripple filter.

Experimental validation

To study and validate the simulation results of this proposed work, an experimental setup is created different parameters are evaluated from the experimental study, and a comparison is made between experimental and simulation results. To illustrate the system's capabilities, a hardware prototype is created. It is made up of an AMETEK-made solar array simulator, a step-up converter, a SEMIKRON-made VSC, a CP1104 dSPACE board controller, a BLDC motor, and a DC generator. The combination of a BLDC motor and a resistive load acts as a water pump. Power in this load is inversely related to the square of speed. The power fluctuates concerning speed for the centrifugal pump and the aforementioned loading setup. Therefore, analysis and validation of the system using the selected loading configuration also show that the centrifugal pump load operates successfully. A 180 V, 50 Hz single phase grid supply supports the tests' single pole, and a 1.3 KW BLDC motor with 3000 rpm @ 270 V DC, which is powered by a solar simulator with a 1500 Wp setting is used. The power quality indices are recorded utilizing a power analyzer. The appendices provide the single-phase grid, motor, and PV array specifications. In the upcoming sections, the experiment results of the proposed topology are reviewed.

The existence of PV supply only to the BLDC motor pump

During full radiation, maximum power can be extracted from PV as illustrated in Fig. 12. Hence, the motor runs at its maximum speed and load as the array supplies 1.5 kW. This situation does not necessitate the use of grid power. The tracking efficiency is 99.7%.

Existence of grid supply only to BLDC motor pump

When it's necessary to pump water at night, the utility grid provides all of the required power. Figure 13a illustrates the drawing of an in-phase supply current that is equal to 8.15 A about the supply voltage, V_s . A full supply of water is carried as a result of the motor operating continuously using current and speed ratings, as illustrated in Fig. 13b. The Fig. 13c harmonic spectra attest to the utility's enhanced power quality. The current has a 3.7% THD and is received at a unity power factor. Additionally, the supply voltage's THD is 2.2%. As a result, the recorded waveforms attest to an improvement in power quality following IEEE-519. Compliance with the IEEE-519 standard was verified through extensive simulations and harmonic analysis of the output waveforms. We employed techniques to minimize total harmonic distortion (THD) in the grid interactions, and

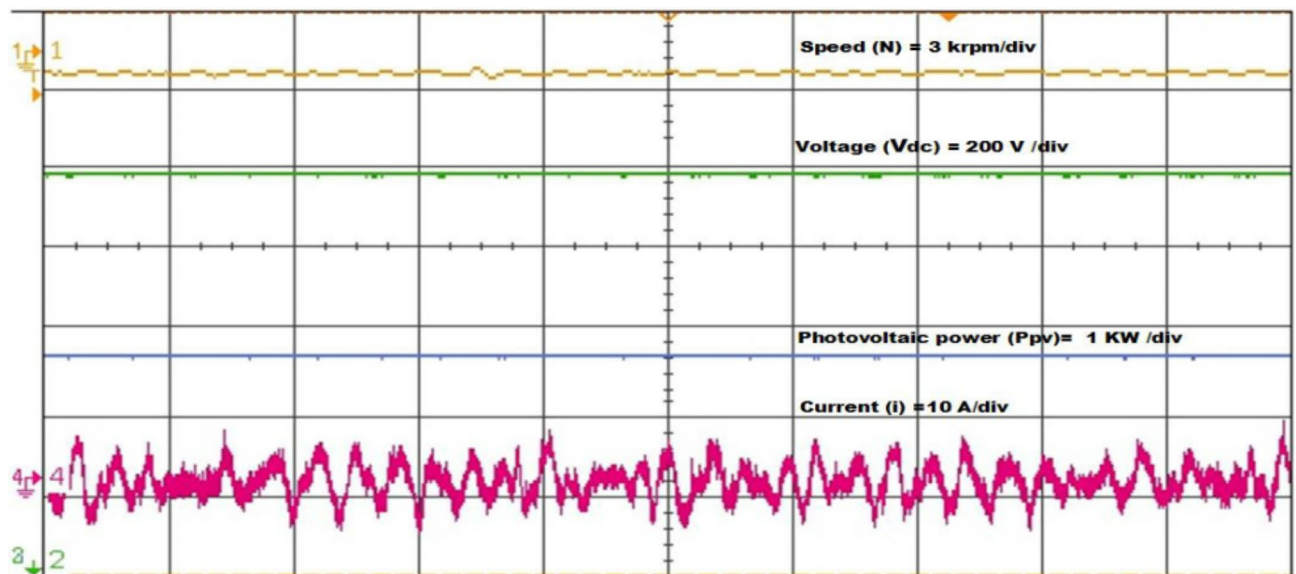
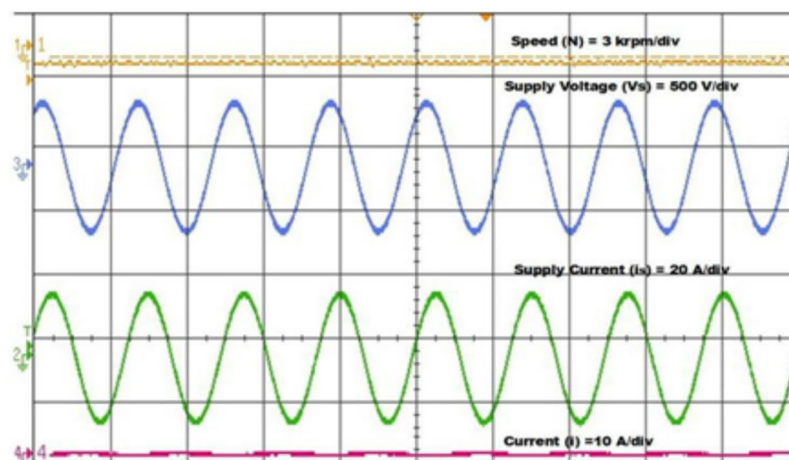
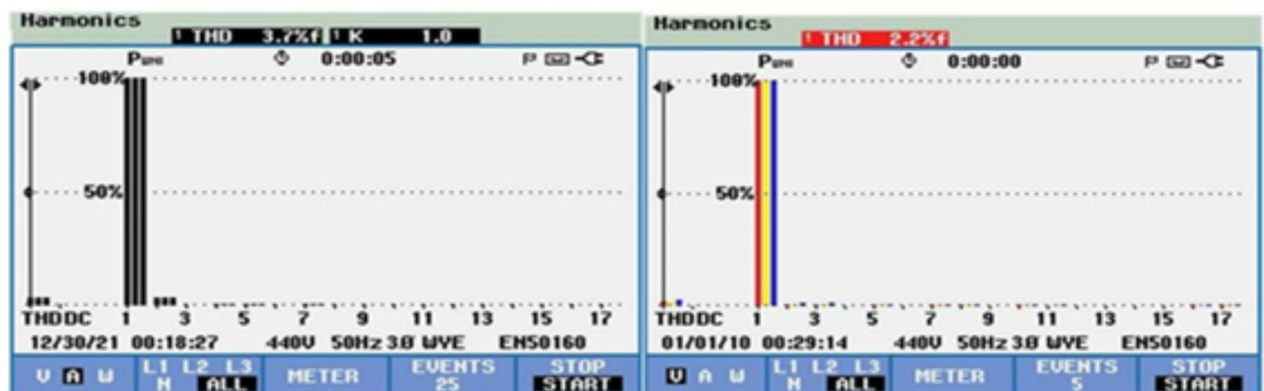


Fig. 12. Steady-state response of pumping system supplied by PV in which BLDC motor operation remains simple.



(a)



(b)

(c)

Fig. 13. Performance at a steady state when the motor is only fed by the grid, as measured by (a) the grid index (b) the grid and motor-pump index (c) the grid's power quality.

our results indicated that THD levels were well within acceptable limits, ensuring that our system maintains high power quality.

Grid is fed by PV array

In this method, the solar PV power feeds to the load and if excess power is available, it is fed to the grid. As said earlier, when it's not essential to pump water, a PV supplies power to the grid. For instance, 1000 W/m^2 radiation is considered to be available. PV indices at 1000 W/m^2 are shown in Fig. 14a from which it is observed that the 1.5 kWp of power is produced from the PV array.

According to Fig. 14b, this power is delivered to the grid. Though the voltage and current of the DC bus are out of phase, the bus voltage is still maintained at 270 V . Figure 14c also displays a supply voltage and current. For this situation, the power quality metrics are also displayed as illustrated in Fig. 14d. These indicators attest to the enhanced PQ at AC mains following the IEEE-519 standard. The power produced by a PV decreases as the radiation level does. But whatever power is produced following that certain radiation level is successfully sent into the system. Figure 13 depicts this situation in detail. The PV array generates 1050 Wp , which corresponds to a radiation level of 700 W/m^2 , as shown in Fig. 15a. According to Fig. 15b, this quantity of power is delivered

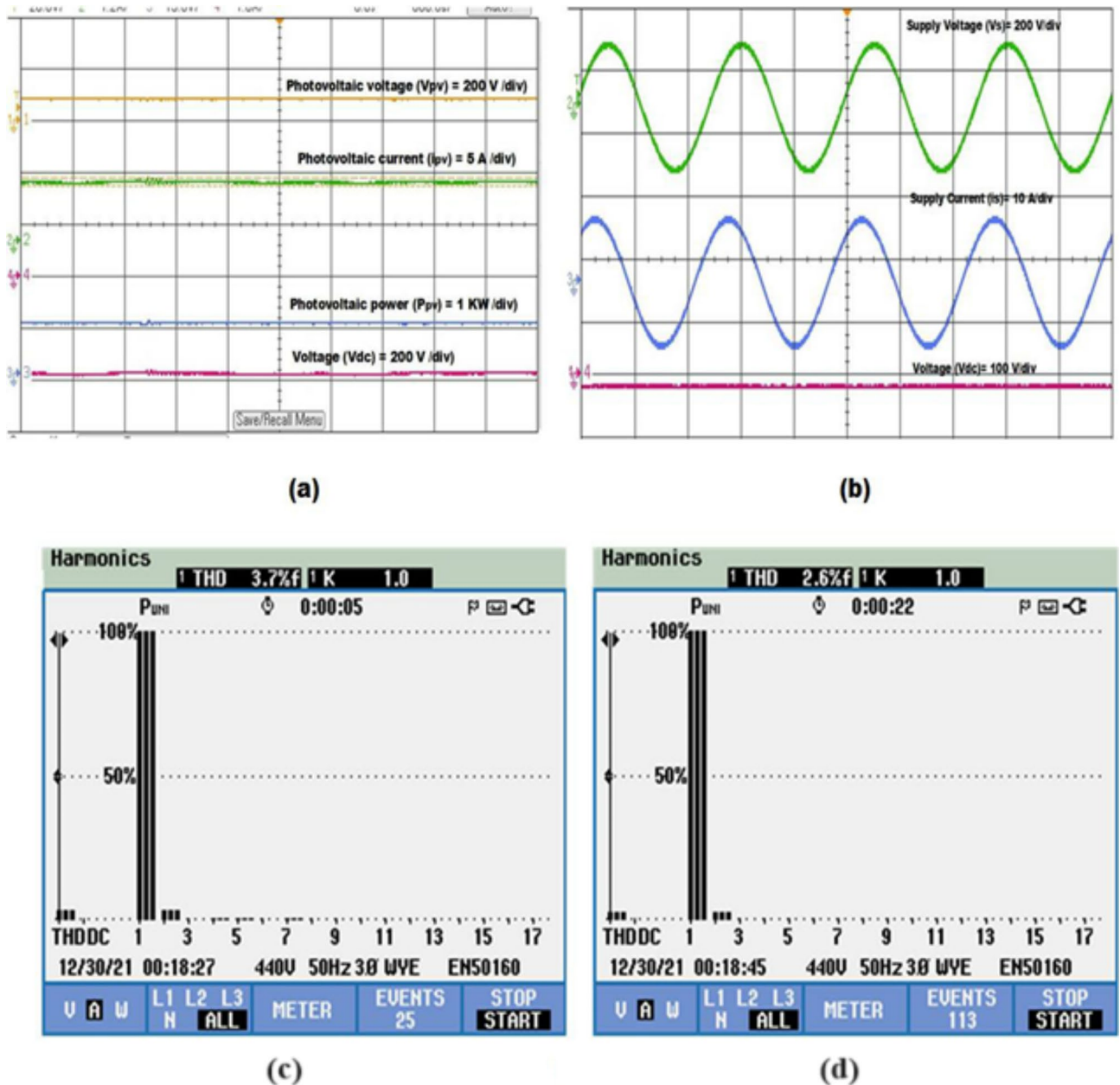


Fig. 14. (a) Steady-state performance of PV during 1000 W/m^2 , (b) Steady-state performance of grid during 1000 W/m^2 , (c) PQ of the grid during 1000 W/m^2 , (d) PQ when the grid is fed by PV at 1000 W/m^2 .

to the grid. The power is fed with a power factor of unity, and the V_{dc} is continually controlled at 270 V. To guarantee enhanced power quality at a single-phase grid, the different grid indices are in Fig. 15c.

To demonstrate its viability in dynamic circumstances, the suggested system must react appropriately to anything that changes the weather profile. In Fig. 15, the system reactions to dynamically changing radiation levels are depicted. The PV response is expressed in Fig. 16a, which shows a step change in the radiation level from 1000 W/m^2 to 700 W/m^2 . A 700 W/m^2 reduction in generated power occurs. As a result, Fig. 16b grid current shows how electricity provided to the system rapidly decreases. The 270 Vdc is controlled by the power flow control.

Grid and PV arrays both supply the BLDC motor-pump

Insufficient PV electricity generation during poor weather is enough to operate a water pump at maximum capacity. Grid support for utilities is required in that circumstance. The proposed PFC calculates the energy usage that must be taken from the grid to fully supply the motor pump. Radiation of 400 W/m^2 will be available at some point to illustrate this instance. Figure 17a depicts a steady-state reaction to this radiation. The PV array can only generate 605 Wp. The motor nevertheless reaches its rated speed and draws the rated current. Drawing the necessary amount of power from the grid enables this. The utility draws 4.85 A (rms) from the system. In the end, the pump disperses all of the water. The power factor, grid indices, and THD of the grid along with grid quality are shown in Fig. 17b. The IEEE-519 standard allows for acceptable THDs for grid voltage and current. Additionally, 0.89 kVA of required power is consumed in Fig. 16b, along with a rms current (4.85 A).

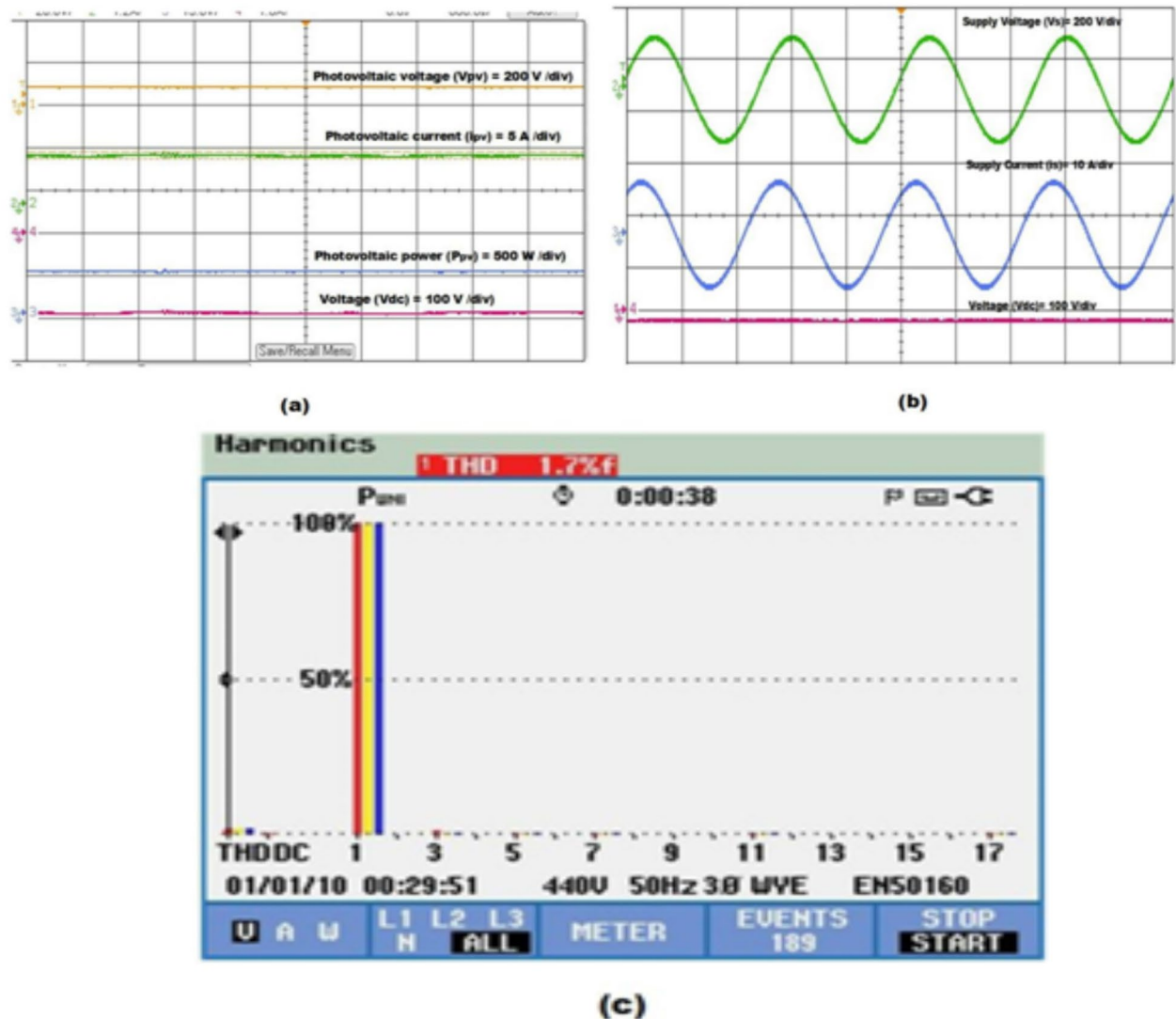
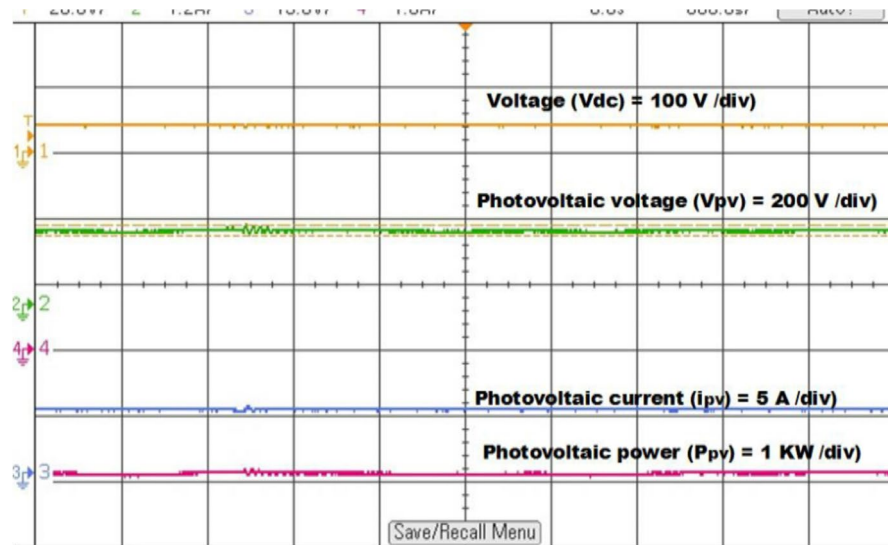
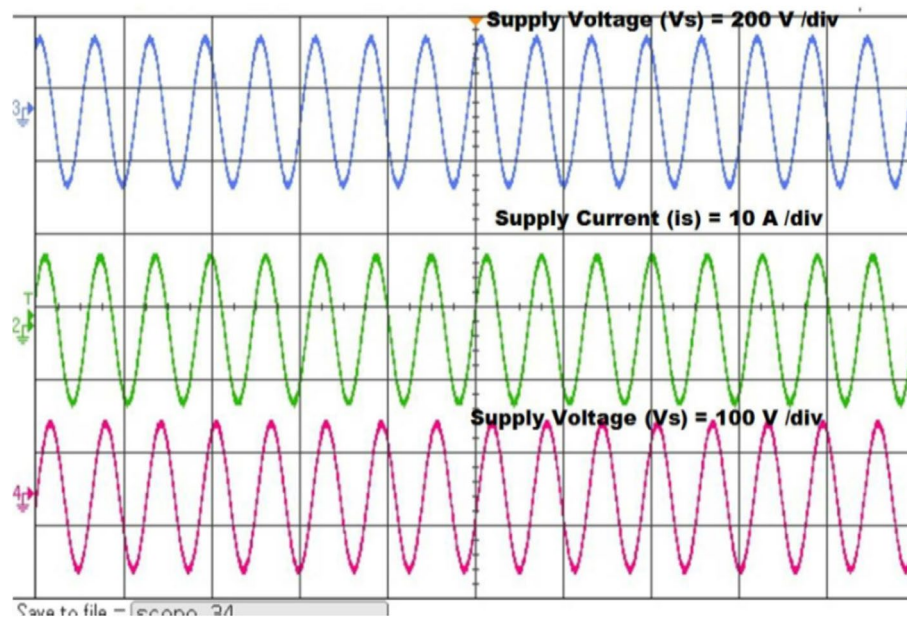


Fig. 15. (a) Steady-state performance of PV during 400 W/m^2 , (b) Steady-state performance of grid during 400 W/m^2 , (c) Power quality of grid during 400 W/m^2 .



(a)



(b)

Fig. 16. (a) Dynamic performance of PV during variation of radiation level from 1000 W/m^2 to 700 W/m^2 and (b) Dynamic performance of grid during respective change in variation.

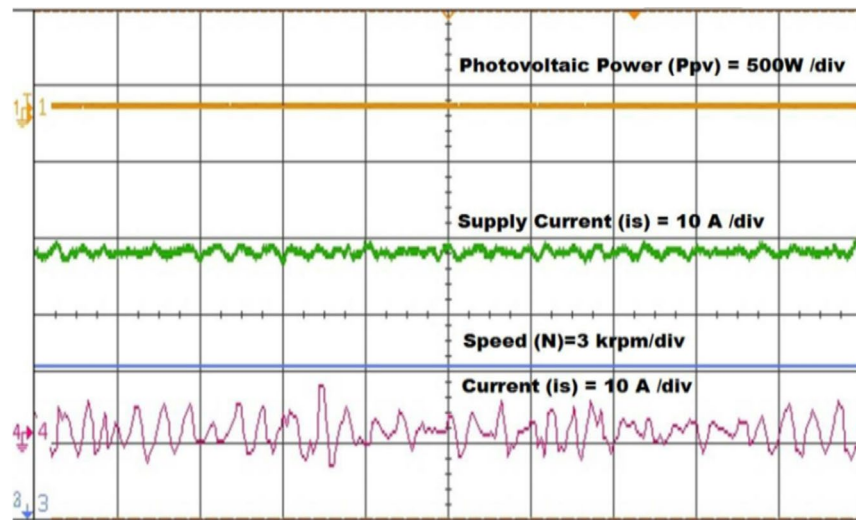
Grid failure condition

When the utility grid delivers a portion of the motor-power pump while it is operating, the proposed system turns into a stand-alone PV-water pumping system if it fails. Even if the motor pump is no longer operating at its peak efficiency, the system still works effectively. The reaction of the system during dynamic grid failure is shown in Fig. 18. The pumping system is running at 700 W/m^2 just before the grid outage. With the assistance of the utility grid, it is run at its rated condition. The amount of electricity used from the grid is eliminated when the grid falls, which reduces the motor speed and current by 700 W/m^2 . Moreover, the system is still functional in independent operation and offers a water supply comparable to a 700 W/m^2 PV power that is readily available.

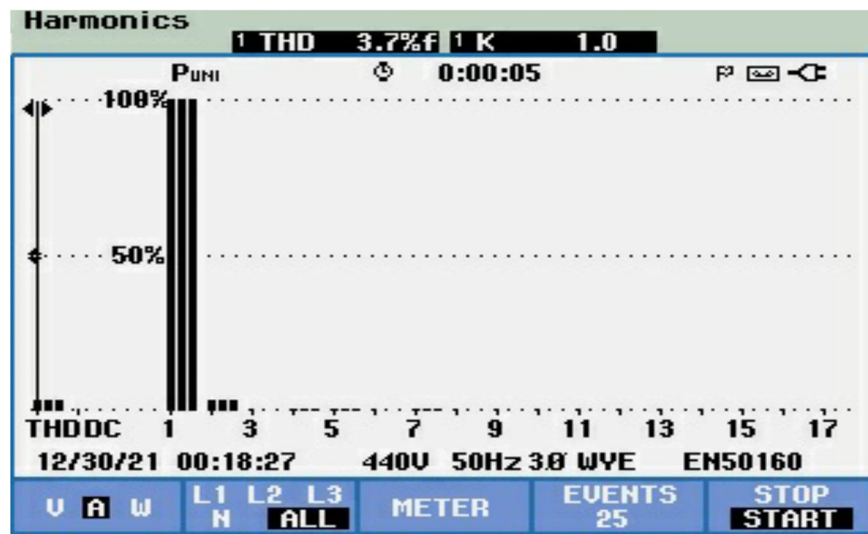
The results from the experiment are very comparable to those from the simulation. This establishes the simulation model's authenticity. Based on several operating factors it decided that the quantity of power travels either from grid and PV to pump or between PV and grid. Table 3 displays the power distribution due to differential solar radiations to carry out the specified function.

Table 4 describes the behavior of the aforementioned controllers about PQ.

Table 5 defines the RMS voltages from various controllers. It demonstrates that the PI controller attains a significant voltage THD of 3.736. The PI controller, on the other hand, managed to achieve a load voltage THD



(a)



(b)

Fig. 17. Performance during steady conditions when combined PV and grid supplies pump, as shown in (a) power supply to pump by combined grid and PV (b) Quality of grid power.

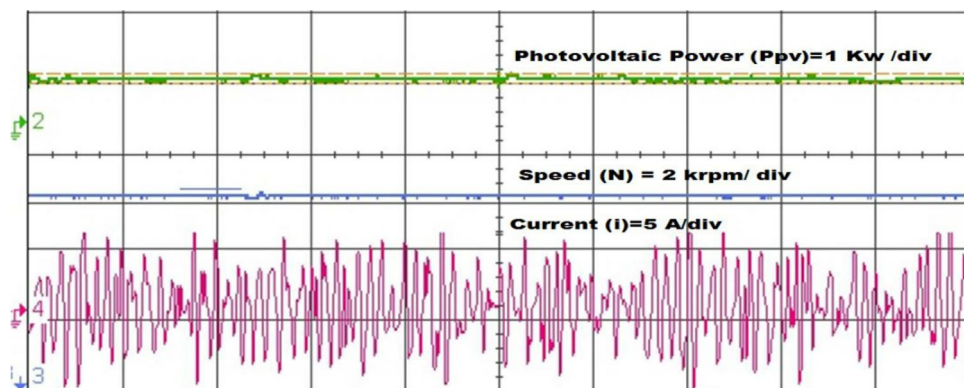


Fig. 18. Dynamic performance under grid failure.

Solar radiation (W/m ²)	Water pumping is not needed	Water pumping is needed	
	Power fed to grid (W)	Power drawn from grid (W)	Power drawn from PV array (W)
0	0	1470	0
100	105	1335	144
200	234	1188	294
300	365	1040	449
400	501	890	605
500	634	740	747
600	765	592	897
700	890	444	1046
800	1030	301	1195
900	1167	150	1348
1000	1300	0	1500

Table 3. Varied operational conditions for the distribution of power.

Solution techniques	Fault time (sec)	Source voltage during a fault (V)	Sag amplitude due to fault (V)	Load voltage (V)	THD (%)
ANFIS	0.04–0.08	400	200	390	1.7
PI	0.04–0.08	400	200	380	3.7

Table 4. Performance evaluation of the various methods.

Time instant in sec	RMS voltage (V)	RMS voltage (Volts) after PQ enhancement	
		ANFIS	PI
0.025	90	73	42
0.15	75	105.36	82

Table 5. ANFIS and PI-based controllers’ RMS voltages.

of 2.629%. However, the ANFIS method, whose value is 1.739%, is discovered to have a lower THD than all remotes.

The PQ has increased by 76% as a result of the PI and ANFIS compensating for voltage and current disturbances that occurred at improper times. The ANFIS technique then successfully addresses the drawbacks of the aforementioned techniques, increases the PQ to a whopping 98%, and provides a better reduction of THD than other methods.

Conclusion

In this paper, the application of BLDC motor drive for a single-phase grid-interactive water pumping system has been proposed. Whatever the weather condition is, the VSC’s bidirectional power flow control allows for greater water pumping capability and resource utilization. The power flow has been adjusted using a straightforward UVT-generating approach. According to the IEEE-519 standard, all power quality requirements have been considered. Without the need for any current sensing components, BLDC motor pump speed control has been accomplished. By lowering switching losses, VSI’s basic frequency switching has improved the efficiency of the entire system. The proposed method is a dependable water pumping system and a method of making money by charging the utility for electricity while water pumping is not necessary. However, due to incorrectly chosen PI control gains, the stability of the system is impacted, which is remedied by utilizing the suggested intelligent fuzzy logic controller. It demonstrates that the PI controller attains a significant voltage THD of 3.736. The PI controller, on the other hand, managed to achieve a load voltage THD of 2.629%. However, the ANFIS method, whose value is 1.739%, is discovered to have a lower THD than all remotes with improved features, it lessens abrupt swings while maintaining steady DC-link voltage. Controller Tuning and Stability, Environmental Variability, Cost Considerations, Grid Interaction Limitations, and Scalability are the challenges faced during this research work.

Scope for future work

Future work will focus on enhancing the robustness of the system under extreme weather conditions and integrating advanced energy storage solutions to further improve reliability and efficiency. Additionally, we aim to conduct field tests in diverse geographical locations to validate the system’s adaptability and performance.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Received: 21 September 2024; Accepted: 25 October 2024

Published online: 29 October 2024

References

1. Nisha, R. & Sheela, K. G. Review of PV fed water pumping systems using BLDC Motor. *Mater. Today Proc.* **24**, 1874–1881 (2020).
2. Kumar, R. & Singh, B. Grid interactive solar PV-based water pumping using BLDC motor drive. *IEEE Trans. Ind. Appl.* **55**(5), 5153–5165 (2019).
3. CH, H. B., Dhanamjayulu, C., Kamwa, I. & Muyeen, S. M. A novel on intelligent energy control strategy for micro grids with renewables and EVs. *Energy Strategy Reviews.* **52**, 101306 (2024).
4. Zhang, H. et al. High-dynamic and low-cost sensorless control method of high-speed brushless DC motor. *IEEE Trans. Industr. Inf.* **19**(4), 5576–5584 (2022).
5. Talbi, B. et al. A high-performance control scheme for photovoltaic pumping system under sudden irradiance and load changes. *Sol. Energy.* **159**, 353–368 (2018).
6. Zhao, D., Cui, L. & Liu, D. Bearing weak fault feature extraction under time-varying speed conditions based on frequency matching demodulation transform. *IEEE ASME Trans. Mechatron.* **28**(3), 1627–1637 (2022).
7. Gao, J. et al. Design and optimization of a novel double-layer Helmholtz coil for wirelessly powering a capsule robot. *IEEE Trans. Power Electron.* (2023).
8. Khan, M. T. A., Norris, G., Chattopadhyay, R., Husain, I. & Bhattacharya, S. Autoinspection and permitting with a PV utility interface (PUI) for residential plug-and-play solar photovoltaic unit. *IEEE Trans. Ind. Appl.* **53**(2), 1337–1346 (2016).
9. Mathankumar, M., Viswanathan, T., Balachander, K. & Suryaprakash, S. Design and implementation of improved sliding mode control for electric vehicle voltage stabilization. *Mater. Today Proc.* **45**, 1747–1749 (2021).
10. Mathankumar, M., Pozhilan, S., Pujakaleeswari, N., Sivanithish, R. K. & Manoj, C. An efficient closed loop control of PV system with MPPT for constant load applications. In *2023 2nd International Conference on Advancements in Electrical, Electronics, Communication, Computing and Automation (ICAECA)* 1–4 (IEEE, 2023).
11. Aashoor, F. A. O. & Robinson, F. V. P. Maximum power point tracking of PV water pumping system using artificial neural based control 1–6 (2014).
12. Wong, Y. S., Chen, J. F., Liu, K. B. & Hsieh, Y. P. High conversion ratio DC-DC converter with isolated transformer and switched-clamp capacitor for Taiwan photon source. *J. Instrum.* **12**(12), T12005 (2017).
13. Sridhar, R., Vishnuram, P. & Sattianadan, D. Efficient single stage photovoltaic pumping system using BLDC motor with grid power export. *J. Sol. Energy Eng.* **141**(5) (2019).
14. Bharathidasan, M. & Indragandhi, V. Design and implementations of high step-up non-isolated DC-DC converters for electric vehicles application. *Int. J. Circuit Theory Appl.* 1–18. (2022).
15. Basha, C. H. & Murali, M. A new design of transformerless, non-isolated, high step-up DC-DC converter with hybrid fuzzy logic MPPT controller. *Int. J. Circuit Theory Appl.* **50**(1), 272–297 (2022).
16. Ma, K., Yang, J. & Liu, P. Relaying-assisted communications for demand response in smart grid: Cost modeling, game strategies, and algorithms. *IEEE J. Sel. Areas Commun.* **38**(1), 48–60 (2019).
17. Rajesh, J. S., Karthikeyan, R. & Revathi, R. Hybrid DPSO based MPPT control of high static gain converter in photovoltaic system for DC microgrid applications. In *IOP Conference Series: Materials Science and Engineering* Vol. 937 012010 (IOP Publishing, 2020).
18. Murshid, S. & Singh, B. Single stage autonomous solar water pumping system using PMSM drive. *IEEE Trans. Ind. Appl.* **56**(4), 3985–3994 (2020).
19. Viswanathan, T., Mathankumar, M., Rajaguru, R. & Sasikumar, C. Combinatorial optimization technique for improving performance of PV modules under partial shading conditions. *Mater. Today Proc.* **45**, 1651–1654 (2021).
20. Balan, G. et al. Performance analysis and enhancement of brain emotion-based intelligent controller and its impact on PMBLDC motor drive for electric vehicle applications. *Energy Sour. Part A Recover. Utilization Environ. Eff.* **44**(3), 5640–5664 (2022).
21. Shukla, S. & Singh, B. Adaptive speed estimation with fuzzy logic control for PV-grid interactive induction motor drive-based water pumping. *IET Power Electron.* **12**(6), 1554–1562 (2019).
22. Bist, V. & Singh, B. PFC Cuk converter-fed BLDC motor drive. *IEEE Trans. Power Electron.* **30**(2), 871–887 (2014).
23. Sharma, A., Gupta, T. N. & Rawat, M. S. Grid connected solar PV fed constant power water pumping system. In *2021 International Conference on Intelligent Technologies (CONIT)* 1–6 (IEEE, 2021).
24. Zhang, J. et al. A novel multiple-medium-AC-port power electronic transformer. *IEEE Trans. Industr. Electron.* (2023).
25. Mathew, T. M. & Rakhee, R. Non-isolated high gain DC-DC converter for PV applications with closed loop control. In *2019 2nd International Conference on Intelligent Computing, Instrumentation and Control Technologies (ICICICT)* Vol. 1, 627–632 (IEEE, 2019).
26. Singh, P. & Gaur, P. Grid interfaced solar water pumping system with improved space vector modulated direct torque control. *Ain Shams Eng. J.* **11**(4), 1149–1162 (2020).
27. Kumar, R. & Singh, B. Grid interfaced solar PV based water pumping using brushless DC motor drive. In *2016 IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES)* 1–6 (IEEE, 2016).
28. Hussaian Basha, C. H. et al. Development of cuckoo search MPPT algorithm for partially shaded solar PV SEPIC converter. In *Soft Computing for Problem Solving: SocProS 2018* Vol. 1 (Springer, 2018).
29. Miaofen et al. Adaptive synchronous demodulation transform with application to analyzing multicomponent signals for machinery fault diagnostics. *Mech. Syst. Signal Process.* **191**, 110208 (2023).
30. El-Samahy, A. A. & Shamseldin, M. A. Brushless DC motor tracking control using self-tuning fuzzy PID control and model reference adaptive control. *Ain Shams Eng. J.* **9**(3), 341–352 (2018).
31. Wang, Z. et al. Permanent magnet-based superficial flow velocimeter with ultra low output drift. *IEEE Trans. Instrum. Meas.* (2023).
32. Li, Q. et al. Geospatial analysis of scour development in offshore wind farms. *Mar. Georesources Geotechnology* 1–20 (2024).
33. Wang, T. et al. Rolling element bearing fault diagnosis via fault characteristic order (FCO) analysis. *Mech. Syst. Signal Process.* **45**(1), 139–153 (2014).
34. Li, L. et al. Seasonal hydrogen energy storage sizing: Two-stage economic-safety optimization for integrated energy systems in northwest China. *iScience.* **27**(9) (2024).
35. Oshaba, A. S., Elazim, A., Ali, E. S. & S. M., & MPPT control design of PV generator powered DC motor-pump system based on artificial bee colony algorithm. *J. Electr. Eng.* **14**(4), 10–10 (2014).
36. Hussaian Basha, C. H. & Rani, C. Performance analysis of MPPT techniques for dynamic irradiation condition of solar PV. *Int. J. Fuzzy Syst.* **22**(8), 2577–2598 (2020).
37. Chandra, S., Gaur, P. & Pathak, D. Radial basis function neural network based maximum power point tracking for photovoltaic brushless DC motor connected water pumping system. *Comput. Electr. Eng.* **86**, 106730 (2020).

38. Mishra, A. K. & Singh, B. High gain single ended primary inductor converter with ripple free input current for solar powered water pumping system utilizing cost-effective maximum power point tracking technique. *IEEE Trans. Ind. Appl.* **55**(6), 6332–6343 (2019).
39. Hang, J. et al. Improved fault diagnosis method for permanent magnet synchronous machine system based on lightweight multi-source information data layer fusion. *IEEE Trans. Power Electron.* (2024).
40. He, W. et al. Robust diagnosis of partial demagnetization fault in PMSMs using radial air-gap flux density under complex working conditions. *IEEE Trans. Industr. Electron.* (2024).
41. Wang, H. et al. A MTPA and flux-weakening curve identification method based on physics-informed network without calibration. *IEEE Trans. Power Electron.* (2023).
42. Shirkhani, M. et al. A review on microgrid decentralized energy/voltage control structures and methods. *Energy Rep.* **10**, 368–380 (2023).
43. Rong, Q. et al. Virtual external perturbation-based impedance measurement of grid-connected converter. *IEEE Trans. Industr. Electron.* (2024).
44. Mohammed, O. A. et al. Solar PV based super lift LUO converter for BLDC motor drive. In *Journal of Physics: Conference Series* Vol. 1916, 012144 (IOP Publishing, 2021).
45. Rong, Q. et al. Asymmetric sampling disturbance-based universal impedance measurement method for converters. *IEEE Trans. Power Electron.* (2024).
46. Ma, Y. et al. Optimized design of demagnetization control for DFIG-based wind turbines to enhance transient stability during weak grid faults. *IEEE Trans. Power Electron.* (2024).
47. Li, P. et al. A distributed economic dispatch strategy for power–water networks. *IEEE Trans. Control Netw. Syst.* **9**(1), 356–366 (2021).
48. Basha, C. H. & Rani, C. Different conventional and soft computing MPPT techniques for solar PV systems with high step-up boost converters: A comprehensive analysis. *Energies*. **13**(2), 371 (2020).
49. Kumar, R. & Singh, B. BLDC motor-driven solar PV array-fed water pumping system employing zeta converter. *IEEE Trans. Ind. Appl.* **52**(3), 2315–2322 (2016).
50. Lu, W., Han, J., Li, S. & Lu, H. H. C. Mitigating line frequency instability of boost PFC converter under proportional outer-voltage loop with additional third current-harmonic feedforward compensation. *IEEE Trans. Circuits Syst. I Regul. Pap.* **66**(11), 4528–4541 (2019).
51. Duan, Y., Zhao, Y. & Jiangping, H. An initialization-free distributed algorithm for dynamic economic dispatch problems in microgrid: Modeling, optimization and analysis. *Sustainable Energy Grids Networks*. **34**, 101004 (2023).
52. Park, M. H., Baek, J., Jeong, Y. & Moon, G. W. An interleaved totem-pole bridgeless boost PFC converter with soft-switching capability adopting phase-shifting control. *IEEE Trans. Power Electron.* **34**(11), 10610–10618 (2019).
53. Su, B. & Lu, Z. An interleaved totem-pole boost bridgeless rectifier with reduced reverse-recovery problems for power factor correction. *IEEE Trans. Power Electron.* **25**(6), 1406–1415 (2010).
54. Nguyen, H. V. & Lee, D. C. Reducing the dc-link capacitance: A bridgeless PFC boost rectifier that reduces the second-order power ripple at the dc output. *IEEE Ind. Appl. Mag.* **24**(2), 23–34 (2018).
55. Jacoboski, M. J., de Bastiani Lange, A. & Heldwein, M. L. Closed-form solution for core loss calculation in single-phase bridgeless PFC rectifiers based on the iGSE method. *IEEE Trans. Power Electron.* **33**(6), 4599–4604 (2017).
56. Lamo, P., Lopez, F., Pigazo, A. & Azcondo, F. J. Stability and performance assessment of single-phase T/4 PLLs with secondary control path in current sensorless bridgeless PFCs. *IEEE J. Emerg. Sel. Top. Power Electron.* **6**(2), 674–685 (2018).
57. Alam, M. et al. A hybrid resonant pulse-width modulation bridgeless AC–DC power factor correction converter. *IEEE Trans. Ind. Appl.* **53**(2), 1406–1415 (2016).
58. Riordan, C., Hulstrom, R., Cannon, T. & Myers, D. Solar radiation research for photovoltaic applications. *Solar Cells*. **30**(1–4), 489–500 (1991).
59. Voyant, C. et al. Machine learning methods for solar radiation forecasting: A review. *Renew. Energy*. **105**, 569–582 (2017).
60. Ibrahim, I. A. & Khatib, T. A novel hybrid model for hourly global solar radiation prediction using random forests technique and firefly algorithm. *Energy. Conv. Manag.* **138**, 413–425 (2017).
61. Larson E, V. Forecasting solar irradiance with numerical weather prediction models. *Solar energy Forecast. resource Assess.* **299**, 318 (2013).
62. Hedar, A. R., Wang, J. & Fukushima, M. Tabu search for attribute reduction in rough set theory. *Soft. Comput.* **12**(9), 909–918 (2008).
63. Kavitha, M. et al. An advanced rice plant disease classification through a modified efficient net deep learning model. In *2024 Second International Conference on Networks, Multimedia and Information Technology (NMITCON)* (IEEE, 2024).
64. Premalatha, S., Dash, S. S. & Babu, P. C. Power quality improvement features for a distributed generation system using shunt active power filter. *Procedia Eng.* **64**, 265–274 (2013).
65. Bhargavi, K. M. et al. A novel development of advanced control approach for battery-fed electric vehicle systems. *Sci. Rep.* **14**(1), 20194 (2024).
66. Ahmed, W., Manohar, P. & Hussaian Basha, C. H. A novel transient analysis of multiterminal VSC-HVDC system incorporating superconducting fault current limiter. *International Transactions on Electrical Energy Systems*. **1**, 5549066. (2024).
67. Ashwini Kumari, P. et al. Adaptive RAO ensemble dichotomy technique for the accurate parameters extraction of solar PV system. *Sci. Rep.* **14**(1), 12920 (2024).
68. Bhardwaj, M. & Choudury, S. *Digitally Controlled Solar Micro Inverter Design using C2000 Piccolo Microcontroller* (Texas Instrument, 2014).
69. Zakzouk, N. E., Abdelsalam, A. K., Helal, A. A. & Williams, B. W. PV single-phase grid-connected converter: DC-link voltage sensorless prospective. *IEEE J. Emerg. Sel. Top. Power Electron.* **5**(1), 526–546 (2016).
70. Hussaian Basha, C. H. & Alsaif, F. A novel development of wide voltage supply DC–DC converter for fuel stack application with PSO-ANFIS MPPT controller. *Sci. Rep.* **14**(1), 18826 (2024).
71. Rao, G. M. & Sankar, R. R. Fuzzy-PI control of grid interact three-phase voltage source inverter. *Int. J. Electr. Eng.* **10**(1), 57–70 (2017).
72. Kouloumpis, V. & Azapagic, A. Integrated life cycle sustainability assessment using fuzzy inference: A novel FELICITA model. *Sustainable Prod. Consum.* **15**, 25–34 (2018).
73. Mosalam, H. A., Amer, R. A. & Morsy, G. A. Fuzzy logic control for a grid-connected PV array through Z-source-inverter using maximum constant boost control method. *Ain Shams Eng. J.* **9** (4), 2931–2941 (2018).
74. Velpula, S. et al. Impact of DFIM controller parameters on SSR characteristics of wind energy conversion system with series capacitor compensation. In *International Conference on Computer Vision and Robotics* (Springer Nature Singapore, 2023).
75. Touti, E. et al. A novel design and analysis adaptive hybrid ANFIS MPPT controller for PEMFC-fed EV systems. *International Transactions on Electrical Energy Systems*. **1**, 5541124 (2024).
76. Mariprasath, T. et al. A novel on high voltage gain boost converter with cuckoo search optimization based MPPTController for solar PV system. *Sci. Rep.* **14**(1), 8545 (2024).
77. Krishnaram, K., Suresh Padmanabhan, T., Alsaif, F. & Senthilkumar, S. Development of grey wolf optimization based modified fast terminal sliding mode controller for PV system with three phase interleaved boost converter. *Scientific Reports*. <https://doi.org/10.1038/s41598-024-59900-z> (2024).

78. Basha, C. H. et al. Design of an LPF based slider controller for THD reduction in solar PV B-4 inverter. In *2019 IEEE International Conference on Electrical, Computer and Communication Technologies (ICECCT)* (IEEE, 2019).
79. Murali, M. et al. Performance analysis of different types of solar photovoltaic cell techniques using MATLAB/simulink. In *Proceedings of Fourth International Conference on Inventive Material Science Applications: ICIMA 2021* (Springer, 2022).
80. Puppala, R. et al. Framework for smart grid to implement a price elasticity-based peak time rebate demand response program. *Front. Energy Res.* **10**, 1079695 (2023).
81. Kumbhar, A. et al. Reducing grid dependency and operating cost of micro grids with effective coordination of renewable and electric vehicle's storage. In *Soft Computing for Problem Solving: Proceedings of the SocProS 2022* 639–653 (Springer Nature, 2023).
82. Krishnaram, K., Suresh Padmanabhan, T., Alsaif, F. & Senthilkumar, S. Performance optimization of interleaved boost converter with ANN supported adaptable stepped-scaled P&O based MPPT for solar powered applications. *Scientific Reports*. <https://doi.org/10.1038/s41598-024-58852-8> (2024).
83. Crystalline Silicon, P. V. Modules. Available: <https://www.platendt.nl/pdf/bisol-0906-en.pdf>.
84. Mohan, N., Undeland, T. M. & Robbins, W. P. *Power Electronics: Converters, Applications, and Design* (John Wiley & Sons, 2003).
85. Singh, B., Bist, V., Chandra, A. & Al-Haddad, K. Power factor correction in bridgeless-Luo converter-fed BLDC motor drive. *IEEE Trans. Ind. Appl.* **51**(2), 1179–1188 (2014).
86. Kumar Agarwal, R., Hussain, I. & Singh, B. Three-phase single-stage grid tied solar PV ECS using PLL-less fast CTF control technique. *IET Power Electron.* **10**(2), 178–188 (2017).

Author contributions

All the authors contributed to this research work in terms of concept creation, conduct of the research work, and manuscript preparation.

Funding

No funding was received for this research work.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to J.S.R.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2024